EqRAG

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Abstract

This report explores the application of Large Language Models (LLMs) to the task of financial fundamental analysis and stock price movement prediction. We leverage the FinGPT-Forecaster dataset to evaluate the performance of Qwen-2.5-Math-7B and DeepSeek-Math-7B-Instruct. We compare In-Context Learning (ICL) against Finetuning strategies, specifically focusing on Parameter-Efficient Finetuning (PEFT) methods like LoRA and IA3. Our results demonstrate that while ICL struggles with this complex reasoning task, finetuned models, particularly Qwen-2.5 with LoRA achieve significantly higher accuracy (51.7%). Furthermore, we investigate the impact of LoRA rank variations and compare PEFT against Full Finetuning.

1 Picking a Task and Dataset

1.A Task Description & Motivation

We address the task of **Fundamental Stock Analysis**. The goal is to predict the stock price movement (e.g., BUY, SELL, or HOLD) of a specific company for the upcoming week based on a combination of recent news headlines and basic financial metrics (ratios). Additionally, the model must generate a structured analysis listing positive developments, potential concerns, and a summary rationale.

This task is compelling because it requires the model to synthesize heterogeneous information i.e. qualitative textual news and quantitative financial metrics (e.g., P/E ratio, ROE) to perform complex reasoning (Tang et al., 2025; Zhang et al., 2025). Capabilities required include numerical literacy, causality reasoning (linking news to market sentiment), and domain-specific financial knowledge.

1.B Data Description

We utilize the *FinGPT/fingpt-forecaster-dow30-202305-202405* dataset. This dataset focuses on the Dow 30 components over a one-year period.

- **Data Splits:** The dataset was split temporally to prevent look-ahead bias. We utilized a standard train/test split. The train split consisted of 1230 examples and the testing split consisted of 300 distinct prediction examples.
- **Example:** An input consists of a company profile (e.g., American Express), a list of news summaries from the past week, and a dictionary of financial metrics (Asset Turnover, P/E, etc.).

1.C Ethical Considerations

Deploying AI for financial forecasting involves significant ethical risks, including potential financial harm and model bias (Lopez-Lira et al., 2025).

- **Financial Harm:** "Hallucinations" or incorrect predictions can lead to substantial financial loss for users relying on the system for trading.
- **Bias:** The dataset focuses on Dow 30 (large-cap US stocks). The model may not generalize to small-cap or international markets, potentially reinforcing investment biases toward established Western entities.
- **Market Manipulation:** Widespread use of such automated agents could theoretically increase market volatility if they all react similarly to specific news triggers.

1.D Formulation of Training Data

We formulate the task as a **generative instruction-tuning task**.

- **Input:** A structured prompt enclosed in `[INST]` tags containing a system role ("You are a seasoned stock market analyst"), company introduction, news headlines, and financial JSON data.

- **Output:** A structured text response containing: `[Positive Developments]`, `[Potential Concerns]`, and `[Prediction & Analysis]` and `[Action]` (i.e. BUY, SELL or HOLD).

- **Preprocessing:** The dataset did not include the action feature, we synthetically generate the action feature using a rule, such that if the stock is predicted to go up by more than 1% then the action selected is BUY, if the stock is predicted to go down by more than 1%, then the action selected is SELL, or else HOLD. We add this in the output field in the dataset. We also add an instruction in the input to produce this Action in the output.

1.E Method for Evaluation

We assess performance using **Accuracy** and **Macro F1 Score**.

- **Accuracy:** Measures the percentage of times the model correctly predicts the direction of the stock movement (BUY/SELL/HOLD).

- **F1 Macro:** Harmonic mean of precision and recall, ensuring the model performs well across both all classes.

We chose these metrics because stock movement is a classification problem embedded within a generation task. We parse the generated text to extract the prediction action for comparison against ground truth.

2 Adapting a Language Model to your Task

2.A Method for In-Context Learning

For In-Context Learning (ICL), we utilized a **0-shot approach**. Given the complexity of the financial data and the strict output format required, we relied on the model's internal knowledge and instruction-following capabilities without providing few-shot examples, as the prompt length was already substantial.

We selected two math-specialized models, hypothesizing that their numerical reasoning capabilities would aid in interpreting financial ratios (Tang et al., 2025).

1. **DeepSeek-Math-7B-Instruct**

2. **Qwen-2.5-Math-7B**

We did a series of iterations before we selected the prompt structure in Figure 1. First, we instructed the model with the following line: "You are a seasoned stock market analyst.". This alone provides most of the "learning" in the prompt. We also then give further instructions of the task in text, but performance increased when we gave it the answer format/structure in the system prompt, i.e. to break down the positive developments and concerns and then provide a prediction, analysis and an action. This worked because this simulated the Chain of Thought (Wei et al., 2023) process i.e. approaching the problem step-by-step.

Prompt Structure

```
[INST]<<SYS>> You are a seasoned
stock market analyst.  Your
task is to list the positive
developments and potential
concerns for companies based on
relevant news and basic financials
from the past weeks, then provide
an analysis and prediction for the
companies' stock price movement
for the upcoming week.  Your
answer format should be as follows:
[Positive Developments]:  1.  ...
[Potential Concerns]:  1.  ...
[Prediction & Analysis] Prediction:
...  Analysis:  ...  [Action]
BUY/SELL/HOLD:<</SYS>>

[Company Introduction]:  American
Express Co is a leading entity in
the Financial Services sector...

[Headline]:  Credit cards:
issuers could use their own
flexible friends [Summary]:
Concerns profitability has
peaked... [Headline]:  Dow's nearly
175-point fall led by losses...
[Summary]:  ...    [...  Additional
News Items ...]

[Basic Financials]:
assetTurnoverTTM: 0.2614 bookValue:
25992 ...  peTTM: 16.9489 ...
roeTTM: 0.2955 ...

Based on all the information
before 2023-05-14, let's first
analyze the positive developments
and potential concerns for AXP.
Come up with 2-4 most important
factors respectively and keep
them concise.  Most factors should
be inferred from company related
news.  Then make your prediction
of the AXP cryptocurrency price
movement for next week (2023-05-14
to 2023-05-21).  Provide a
summary analysis to support your
prediction.[/INST]
```

Figure 1: The structured prompt used for 0-shot ICL and Finetuning. Note the specific system instructions defining the output format.

2.B Method for Finetuning

We implemented Low-Rank Adaptation (LoRA) for parameter-efficient finetuning (Hu et al., 2021). We selected LoRA to minimize computational overhead while retaining the pre-trained reasoning capabilities.

- **Models:** Same as ICL (DeepSeek-Math-7B-Instruct and Qwen-2.5-Math-7B).
- **Hyperparameters:**
  - **Learning Rate:** $3e^{-4}$
  - **Batch Size:** 4
  - **Epochs:** 1 (to prevent overfitting on the limited dataset)
  - **LoRA Config:** Rank (r) = 16, Alpha = 32, Target Modules = All attention layers (["q_proj", "k_proj", "v_proj", "o_proj"]).

We chose these hyperparameters based on standard convergence practices for 7B models, following prior LoRA studies (Hu et al., 2021), and based on our own experimentation. The cosine learning rate scheduler was used to ensure smooth convergence.

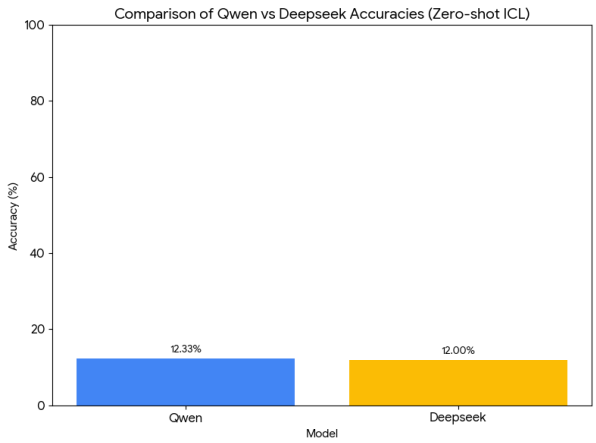

Figure 2: Comparison of Qwen and Deepseek model using ICL

3 Experiments

3.A Results for In-Context Learning

Table 1 presents the 0-shot performance. Both models performed poorly, achieving near-random or worse-than-random accuracy ($\approx$ 12%), highlighting the challenge of numerical reasoning and domain adaptation in financial tasks (Tang et al., 2025). This suggests that without finetuning, the models struggle to map the complex financial/news inputs to the specific output format or fail to extract the correct directional signal.

Model	Accuracy	F1 Macro
DeepSeek-Math-7B (0-shot)	0.120	0.299
Qwen-2.5-Math-7B (0-shot)	0.123	0.254

Table 1: In-Context Learning (0-shot) Performance.

3.B Results for Finetuning

Finetuning yielded a massive improvement over ICL. Table 2 compares the two models trained with LoRA ($r = 16$). Figure 3 illustrates this comparison.

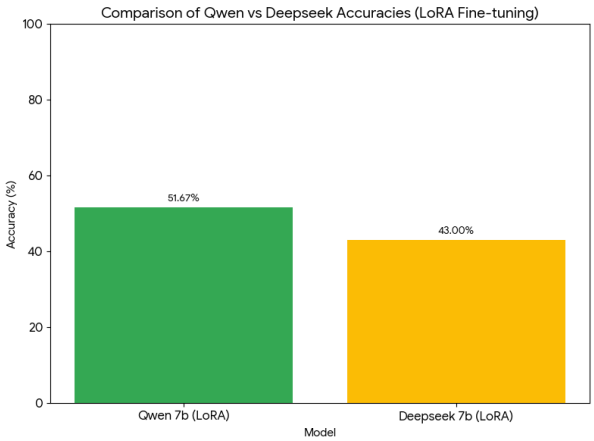

Figure 3: Comparison of Qwen and Deepseek model after finetuning using LoRA

Model	Accuracy	F1 Macro
DeepSeek-Math-7B (LoRA)	0.430	0.510
Qwen-2.5-Math-7B (LoRA)	**0.517**	**0.525**

Table 2: Finetuning Results (LoRA, r=16).

The **Qwen-2.5-Math-7B** model outperformed DeepSeek, achieving an accuracy of 51.7%. While this may appear low compared to NLP tasks, in financial forecasting, an accuracy slightly above 50% is often considered statistically significant in

financial forecasting due to market stochasticity (Lopez-Lira et al., 2025).

3.C Error Analysis

- **Qualitative Analysis:** It is impractical to provide an entire output here due to its length however, we discuss our qualitative findings here. Finetuned models successfully adhered to the JSON-like output structure, whereas ICL models often hallucinated formats, leading to parsing errors (contributing to the 12% accuracy). Further, in ICL the models also often unnecessarily regurgitated the parts of the input, making the answer less coherent.

- **Over/Underfitting:** We trained for only 1 epoch. The training loss decreased steadily, and the validation metrics were stable, suggesting we did not overfit. However, the relatively low accuracy indicates high aleatoric uncertainty in the data itself.

- **Quantitative Analysis:** The accuracy achieved by the Qwen 2.5 Math 7B model with LoRA is 51.7% which much better than choosing randomly between the three options, however, this can be further improved by using a larger base model which is also optimized for reasoning. We also find that using a non-instruction tuned model is better for this task than an instruction tuned model.

3.D Best System for Deployment

Based on the experiments, **Qwen-2.5-Math-7B with LoRA (r=16)** is the best system. It offers the highest accuracy (51.7%) and F1 score (0.525), balancing precision and recall better than the DeepSeek equivalent.

4 Pick your own Experiment

We conducted two additional experiments to deepen our understanding of PEFT strategies.

4.A Experiment 1: FFT vs. LoRA vs. IA3

We compared Full Finetuning (FFT), LoRA (Hu et al., 2021), and IA3(Liu et al., 2022). For FFT, we used a smaller model (Qwen 1.5B) due to compute constraints, compared against Qwen 7B (LoRA/IA3).

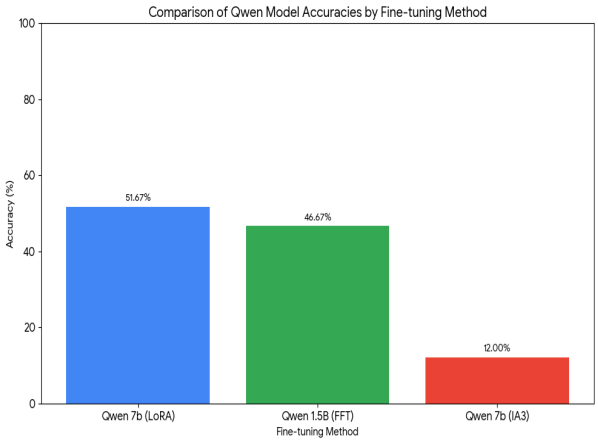

Figure 4: Comparison of Qwen model accuracies using different finetuning methods

Configuration	Accuracy	F1 Macro
Qwen 1.5B (FFT)	0.467	0.483
Qwen 7B (LoRA)	**0.517**	**0.525**
Qwen 7B (IA3)	0.120	0.286

Table 3: Comparison of Tuning Strategies.

Analysis: LoRA on the 7B model outperformed the FFT 1.5B model, consistent with findings that parameter-efficient adaptation often outperforms full tuning on larger pre-trained models. IA3 method failed completely for Qwen (12% accuracy). This suggests that IA3's vector rescaling was insufficient for the domain shift required for this task.

4.B Experiment 2: Varying LoRA Rank

We hypothesized that increasing the LoRA rank (r) would improve expressivity. We tested $r \in \{8, 16, 32, 64\}$ with $\alpha = 2 \times r$.

Rank (r)	Accuracy	F1 Macro
$r = 8$	0.487	0.489
$r = 16$ **(Baseline)**	**0.517**	**0.525**
$r = 32$	0.497	0.518
$r = 64$	0.467	0.507

Table 4: Effect of LoRA Rank on Qwen-7B.

Analysis: As we can observe in figure 5, performance peaked at $r = 16$. Increasing the rank to 32 and 64 resulted in a performance degradation. This suggests that a rank of 16 provides sufficient capacity for the task, and higher ranks may lead to

overfitting on the training noise or require different learning rate scaling.

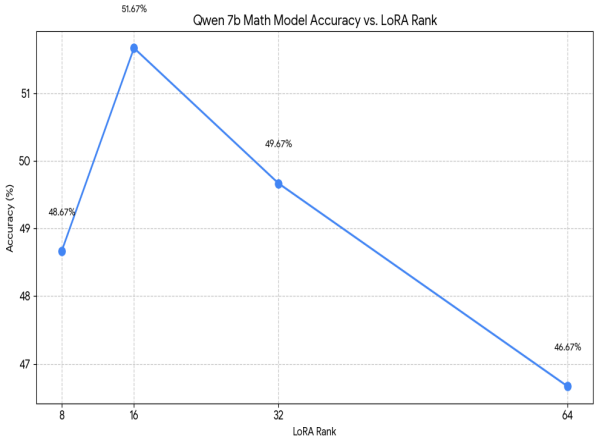

Figure 5: Comparison of Qwen model accuracies using different ranks for LoRA finetuning

Team Member Contributions

Aditya Motwani was responsible for setting up the training pipeline, implementing the LoRA and IA3 configurations, and running the full suite of finetuning experiments (FFT, LoRA variations). He also compiled the experimental results and wrote the Methods and Experiments sections of the report.

Anuj Gupta was responsible for designing and executing the In-Context Learning (ICL) baselines. He curated the prompts used for both ICL and finetuning. Additionally, he contributed to the writing of the Task Description and Ethical Considerations sections of the report.

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